A rank-marked series for the neutral memory field

Actual propagators, Jacobi contraction errors, and finite-time control

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Abstract

We give a uniformly summable representation of the neutral component of the common-coupling complex DSSYK memory field on every fixed real-time interval. Its inputs are highest Gaussian-chaos fields with one output-rank deficit mark. A forced recurrence retains all lower-chaos contractions. The main additional estimate is uniform integrability of the rank-deficit square for the actual exponential, obtained from fixed-degree sixth moments and a graph-norm Taylor tail. This identifies a finite-time computational interface; the limiting marked joint spectral measure remains to be computed.

1 Model and the result to be represented

Use standard CAR, overlapping complex Gaussian p-body couplings, and the same original couplings in every charge sector. The state is the normalized annealed full trace at $\beta = \mu = 0$, with weights $b_m = 2^{-N}\binom{N}{m}$. We take even N, $p \to \infty$, $\lambda_N = p^2/N \to \lambda > 0$, fixed J_s, $\hbar = 1$, and string real time. In sector m write $h = H_m = \sum_\gamma g_\gamma B_\gamma$, $\sum_\gamma B_\gamma^2 = C_m I$; $C_m \le M$ uniformly. Norms mean $\|X\|_{2,m}^2 = \mathbb{E}\,\mathrm{tr}_m\, X^\dagger X$. An independent copy u defines $\Phi(X) = \mathbb{E}_u[uXu]$. The exact energy pinching is $\mathcal{P}_h X = \sum_e P_e X P_e$, including whole degenerate blocks.

Put $\alpha_m = (1 - p/m)_+$ and $\epsilon_m = 1 - \alpha_m^2$ (zero sectors are interpreted directly). The preceding two-component reduction gives $\mathsf{F} = A + B + o_{L^2}(1)$, where the neutral field is

$$B_m(s) = (1 - \mathcal{P}_h) m \epsilon_m e^{-ihs/2} \Phi(e^{i\alpha_m h s}) e^{-ihs/2}. \tag{1}$$

Here A is the charged transfer defect from the previous stage [1, 2]. We retain α_m and $m\epsilon_m/p$ exactly. Our new representation of B is stated on page 6; its construction uses finite-N coefficients throughout.

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2 Exact Jacobi rearrangement, before taking a limit

In a broad central sector $N/4 \le m \le 3N/4$, suppress m and set

$$\mu(r) = \prod_{\ell=0}^{p-1} \left(1 - \frac{r}{m-\ell}\right)_+ \left(1 - \frac{r}{N-m+p-\ell}\right)_+, \qquad \theta_j = \mu(jp), \tag{2}$$

$$\mathfrak{D} = p\mathcal{N} - \mathcal{R}, \qquad Z_j = \mathcal{W}_j(h^j), \qquad U_j = \mathfrak{D} Z_j. \tag{3}$$

$\mathcal{N}$ counts coupling Gaussian chaos; $\mathcal{R}$ is the output $U(N)$ rank. On equivariant functions $\mathfrak{D} \ge 0$, and $\Phi/C = \mu(\mathcal{R})$. These operators commute with one another, but not generally with samplewise multiplication by h. The positive-part extension defines $\mu(jp)$ even beyond the physical rank range.

Let $\lambda_N \in [\lambda_-, \lambda_+]$ and $p \le N/16$. The previous Jacobi construction [2] supplies

$$a_j = \sum_{v<j} \theta_v, \quad 1 \le a_j \le A \ (j \ge 1), \quad A = (1 - e^{-3\lambda_-})^{-1}, \tag{4}$$

$$Q_0 = 1, \quad Q_1 = x, \quad Q_{j+1} = xQ_j - Ca_jQ_{j-1}, \tag{5}$$

$$E_j = Z_j - Q_j(h), \qquad \eta_j = pE_j. \tag{6}$$

Here $a_0 = 0$ and the three-term recurrence starts at $j = 1$. E_j has Gaussian degree at most $j - 2$. The scalar coefficients of the auxiliary bounded Jacobi operator satisfy the entire identity and the bounds

$$e^{ish} = \sum_{j \ge 0} c_j(s) Q_j(h), \qquad |c_j(s)| \le e^{2\sqrt{AM}S} \frac{(2\sqrt{A}S)^j}{j!}, \tag{7}$$

$$\|E_j\|_2 \le \frac{p}{N} L K_*^j \sqrt{j!}\, C^{j/2}, \qquad K_* = 4 + A, \tag{8}$$

$$L = 12\sqrt{2 \cdot 10^6}(\sqrt{\lambda_+} + \lambda_+), \qquad |s| \le S. \tag{9}$$

This is an expansion on the actual Gaussian spectrum, not just on the compact support of an auxiliary measure.

Since $\theta(\mathcal{N}) Z_j = \theta_j Z_j$, exact algebra gives

$$\theta(\mathcal{N}) Q_j(h) = \theta_j Q_j(h) + (\theta_j - \theta(\mathcal{N})) E_j. \tag{10}$$

The sum $\sum c_j \theta_j Q_j(h)$ is a samplewise energy function. All series converge absolutely in annealed L^2. Therefore

$$\boxed{\frac{p}{C}(1 - \mathcal{P}_h)\Phi(e^{ish}) = (1 - \mathcal{P}_h)\left\{ p[\mu(\mathcal{R}) - \theta(\mathcal{N})] e^{ish} + \sum_{j \ge 0} c_j(s)(\theta_j - \theta(\mathcal{N}))\eta_j \right\}.} \tag{11}$$

The second sum cannot be omitted: pE_j is order one on this scale.

3 Fixed-degree higher moments from the existing rank law

The positive rank-product law and the sixth-moment certificate of [4, 3] extend to any fixed number of rank steps. For independent pure ranks a, r, let the output rank be t and $S = a + r - t$. Its Racah identity determines moments through

$$\mathbb{E}\chi_t(n) = \chi_a(n)\chi_r(n), \qquad x_t = t(N + 1 - t), \tag{12}$$

$$F_n = \frac{\chi_a(n)\chi_r(n) - \sum_{j<n} c_{nj}F_j}{c_{nn}}, \quad F_j = \mathbb{E}\prod_{\ell<j}(x_t - x_\ell). \tag{13}$$

Here c_{nj} are the coefficients of $\chi_t(n)$ in that monic polynomial basis; their exact expression and the symbolic certificate are in [4].

The certificate cancels 21 negative Laurent coefficients through order six over the full polynomial ring $\mathbb{Q}[A_0, B_0, z]$, with $A_0 = a/\sqrt{N}$, $B_0 = r/\sqrt{N}$, $z = (m - N/2)/\sqrt{N}$. Its denominators have nonzero limiting constants on compact sets. For any fixed K, X, it consequently applies to

$$a \le p, \quad r \le Kp, \quad |m - N/2| \le Xp. \tag{14}$$

Indeed $x_{a+r} - x_t = S[N + 1 - 2(a + r) + S]$, whose bracket is at least $N/2$ for sufficiently large N at fixed K. Hence $\mathbb{E}S^6$ is uniformly bounded. This extension changes the compact set of the old certificate, not the physical interaction size p.

For a Hamiltonian rank step define $\delta = p + r - t = (p - a) + S$. The input rank deficit $p - a$ has the previously proved factorial tail. Mixture over a and $(x + y)^6 \le 32(x^6 + y^6)$ give finite constants

$$M_{K,6} = \sup_{r\le Kp} \mathbb{E}_r\delta^6 < \infty, \qquad S_{K,4} = \sup_{r\le Kp} \mathbb{E}_r\delta^4 \le M_{K,6}^{2/3}. \tag{15}$$

These constants may depend on $K, X, \lambda_\pm$. The positive Markov chain of k independent inputs has

$$D_k = kp - r_k = \sum_{j<k} \delta_j, \qquad \mathbb{E}D_k^6 \le k^5 \sum_{j<k} M_{j,6}. \tag{16}$$

Gaussian symmetrization is a contraction and commutes with output rank, so

$$\|\mathfrak{D}^a Z_k\|_2^2 \le k! C^k \mathbb{E}D_k^{2a}, \qquad a = 1, 2, 3. \tag{17}$$

This fixed-k conclusion alone does not control the actual exponential. The next page supplies the degree tail before either cutoff is removed.

4 A graph-norm tail for the actual exponential

Let $\mathcal{T}$ denote left multiplication by h. On degree at most r, the all-degree first-commutator bound of [3] reads

$$\|\mathcal{T}\psi\|_2 \le L_H\sqrt{r+1}\|\psi\|_2, \quad L_H = 2\sqrt{M}, \tag{18}$$

$$\|[\mathfrak{D},\mathcal{T}]\psi\|_2 \le K_0(r+1)^{3/2}\|\psi\|_2, \quad K_0 = 2000\sqrt{M(\lambda_+ + \lambda_+^2)}. \tag{19}$$

These hold throughout the broad central interval. Commuting $\mathfrak{D}$ through h^n inductively gives

$$b_n := \|\mathfrak{D}h^n\|_2 \le \frac{K_0 n(n+1)}{2} L_H^{n-1}\sqrt{n!}. \tag{20}$$

For the ordinary Taylor polynomial $P_K(s,h)$, closedness of $\mathfrak{D}$ and the convergent Gaussian moment series imply

$$\sup_{|s|\le S}\|\mathfrak{D}(e^{ish} - P_K(s,h))\|_2 \le T_K(S), \tag{21}$$

$$T_K(S) = \frac{K_0}{2}\sum_{n>K}\frac{n(n+1)S^n L_H^{n-1}}{\sqrt{n!}} \longrightarrow 0. \tag{22}$$

In a compact central charge window, rank orthogonality, creation symmetrization and (15) also give

$$\|[\mathfrak{D},[\mathfrak{D},\mathcal{T}]]\psi_{\le K}\|_2 \le J_K\|\psi\|_2, \qquad J_K = 2\sqrt{M(K+1)S_{K,4}}. \tag{23}$$

The annihilation block is the adjoint of the creation block; no reverse transition probability is assumed. Define $d_0 = 0$ and

$$d_n = L_H\sqrt{n}\, d_{n-1} + 2K_0 n^{3/2} b_{n-1} + J_{n-1}L_H^{n-1}\sqrt{(n-1)!}, \tag{24}$$

$$C_K(S) = \sum_{n\le K} S^n d_n/n!. \tag{25}$$

The identity $\mathfrak{D}^2\mathcal{T} = \mathcal{T}\mathfrak{D}^2 + 2[\mathfrak{D},\mathcal{T}]\mathfrak{D} + [\mathfrak{D},[\mathfrak{D},\mathcal{T}]]$ proves $\|\mathfrak{D}^2 h^n\|_2 \le d_n$. Thus

$$\boxed{\sup_{|s|\le S}\|\mathfrak{D}1_{\mathfrak{D}>R}e^{ish}\|_2 \le T_K(S) + C_K(S)/R.} \tag{26}$$

Choose K first, then R. This is uniform integrability of the $\mathfrak{D}^2$ spectral weight of the complete exponential.

For fixed k,d, differentiating the finite product for μ gives $p[\mu(kp-d)-\mu(kp)] - 4\lambda_N\theta_k d \to 0$. Globally its first term is at most $12\lambda_N d$, so the absolute combined multiplier is bounded by $16\lambda_+\mathfrak{D}$. Degree cutoff (22) and rank cutoff (26) therefore prove

$$p[\mu(\mathcal{R}) - \theta(\mathcal{N})]e^{ish} - 4\lambda_N\theta(\mathcal{N})\mathfrak{D}e^{ish} \longrightarrow 0 \quad \text{in } L^2, \tag{27}$$

uniformly on compact real times and compact central charge windows.

5 Generating every contraction correction from one marked family

Use $\diamond$ for the ordinary coupling-Gaussian Wick product with ordered matrix coefficients, not the spectral q-Wick product. Differentiating the homogeneous polynomial before projecting gives the exact annihilation remainder

$$r_j = \sum_{v<j} [(\Phi - C\theta_v) Z_v] \diamond Z_{j-1-v}, \tag{28}$$

$$\eta_{j+1} = h\eta_j - Ca_j\eta_{j-1} - pr_j, \qquad \eta_0 = \eta_1 = \eta_2 = 0. \tag{29}$$

For a homogeneous degree-v tensor V_v, right creation by h repeated w times proves

$$\|V_v \diamond Z_w\|_2 \le \sqrt{(v+w)!/v!}\, C^{w/2} \|V_v\|_2. \tag{30}$$

One sums $\sum B_\gamma^2 = CI$ before symmetrization at each step. This estimate does not assert a bound for arbitrary products of two matrix-valued Gaussian functions.

Define the finite-parameter source and its forced recurrence by

$$\widehat{r}_j = 4\lambda_N C \sum_{v<j} \theta_v (U_v \diamond Z_{j-1-v}), \tag{31}$$

$$\widehat{\eta}_0 = \widehat{\eta}_1 = 0, \qquad \widehat{\eta}_{j+1} = h\widehat{\eta}_j - Ca_j\widehat{\eta}_{j-1} - \widehat{r}_j. \tag{32}$$

$U_0 = 0$ also gives $\widehat{\eta}_2 = 0$. At fixed j, the slope calculation on Z_v, controlled by (16), yields $\|pr_j - \widehat{r}_j\|_2 \to 0$. Every error has bounded Gaussian degree, so multiplication by h is bounded on it. Induction in (29) gives

$$\|\eta_j - \widehat{\eta}_j\|_2 \longrightarrow 0 \qquad (j \text{ fixed}). \tag{33}$$

The all-degree second-moment bound already established in [3] is

$$\|U_v\|_2 \le \sqrt{v!}\, C^{v/2} \sqrt{K_v}, \qquad K_v = 2 \cdot 10^6 (\lambda_N v^2 + \lambda_N^2 v^4). \tag{34}$$

Set $q_* = e^{-3\lambda_-}$. Since $\theta_v \le q_*^v$, (30) implies $\|\widehat{r}_j\|_2 \le \widehat{L} C^{(j+1)/2} \sqrt{(j-1)!}$, where

$$\widehat{L} = 4\lambda_+ \sqrt{2 \cdot 10^6} \left[\frac{\sqrt{\lambda_+} q_*}{(1-q_*)^2} + \frac{\lambda_+ q_* (1+q_*)}{(1-q_*)^3} \right]. \tag{35}$$

Recurrence (32) now gives the uniform estimate

$$\|\widehat{\eta}_j\|_2 \le \widehat{L} K_*^j C^{j/2} \sqrt{j!}. \tag{36}$$

Equations (8), (7) and (36) give summable tails for both correction series before taking the large-N limit.

6 Closing the actual finite-time representation

It remains to replace $\mathfrak{D}e^{ish}$ by the highest-chaos marked series. Fixed-degree bounds on $\mathfrak{D}^2 Q_j(h)$ follow from the double commutator; those on $\mathfrak{D}^2 Z_j$ follow from (16). Since $E_j = O(1/p)$, spectral Cauchy–Schwarz proves

$$\|\mathfrak{D}E_j\|_2^2 \le \|E_j\|_2 \|\mathfrak{D}^2 E_j\|_2 \longrightarrow 0. \tag{37}$$

For an all-degree bound, the nonadjacent coefficients of Q_j obey $|[x^{j-2\ell}]Q_j| \le C^\ell A^\ell \binom{j-\ell}{\ell}$. Let $R_* = 1 + \sqrt{1+A}$ and $d_+ = \lambda_+ + \lambda_+^2$. Using the first commutator with C in place of M gives

$$\|\mathfrak{D}Q_j(h)\|_2 \le 500\sqrt{d_+}\, j(j+1) R_*^j C^{j/2} \sqrt{j!}. \tag{38}$$

Indeed $\sum_\ell \binom{j-\ell}{\ell} A^\ell 2^{j-2\ell} \le R_*^j$. The Jacobi series therefore converges in the $\mathfrak{D}$ graph norm, and fixed-degree convergence plus its summable tail yields

$$\mathfrak{D}e^{ish} - \sum_{j\ge 0} c_j(s) U_j \longrightarrow 0. \tag{39}$$

Proposition 1. *Define, using only the finite-parameter objects above,*

$$\mathcal{V}_m(s) = \sum_{j\ge 0} c_j(s) \left[4\lambda_N \theta_j U_j + (\theta_j - \theta(\mathcal{N}))\widehat{\eta}_j\right], \tag{40}$$

$$\widetilde{B}_m(s) = \frac{m\epsilon_m}{p} C_m (1 - \mathcal{P}_h) e^{-ihs/2} \mathcal{V}_m(\alpha_m s) e^{-ihs/2}. \tag{41}$$

Set $\widetilde{B}_m = 0$ outside $N/4 \le m \le 3N/4$. For every fixed S, the series is absolutely and uniformly summable, and

$$\sup_{|s|\le S} \sum_m b_m \|B_m(s) - \widetilde{B}_m(s)\|_{2,m}^2 \longrightarrow 0. \tag{42}$$

To prove this, combine (11), (27), (33) and (39); $\theta(\mathcal{N})U_j = \theta_j U_j$. All factors outside the series are bounded: $m\epsilon_m/p \le 2$, $C_m \le M$, real-time phases are unitary, and pinching is contractive. On the central annulus, both fields have uniform bounds by (8), (36) and the previous neutral estimate. The binomial probability of $|m - N/2| > Xp$ is at most $2e^{-2\lambda_- X^2}$; let $X \to \infty$ after N. Outside the broad central interval, $\|B_m\|_2 \le 2pM$ and the probability is at most $2e^{-N/8}$. These bounds establish (42).

7 Low-degree calibration and the joint-kernel task

The exact finite formulas retain conserved terms until the final energy pinching. Define

$$b_1 = (\Phi - C\theta_1)h, \quad b_2 = (\Phi - C\theta_2)Z_2, \quad d_1 = \mathbb{E}\,\mathrm{tr}_m(hb_1). \tag{43}$$

Equivariance makes the contraction of h with b_1 equal to $d_1 I$. The recurrence implies

$$\eta_3 = -pb_1, \tag{44}$$
$$\eta_4 = -p\{h, b_1\} - pb_2 + pd_1 I. \tag{45}$$

The anticommutator itself has scalar contraction $2d_1 I$; the net chaos-zero part of η_4 is $-pd_1 I$. In the central double-scale limit the earlier conventions give

$$\eta_3 = -F/2 - 4\lambda^2 Cq\, h + o_{L^2}(1), \qquad q = e^{-4\lambda}, \quad C = J_s^2/\lambda. \tag{46}$$

The term proportional to h is removed by final pinching, but cannot be dropped inside a subsequent chaos operation. After pinching, the known coefficient calibration is

$$(1 - \mathcal{P}_h)\eta_3 = -F/2 + o_{L^2}(1), \tag{47}$$
$$(1 - \mathcal{P}_h)\eta_4 = -2\lambda C q^2 (Y_- + Y_+) - (1+q)J_h F + o_{L^2}(1), \qquad J_h X = \{h, X\}/2. \tag{48}$$

These relations use the already defined marked fields of [1]. They are coefficient checks; they do not assume interchange of derivatives with an unevaluated limiting time kernel.

Implication for the research program. The previous $\mathsf{F} = A + B + o_{L^2}(1)$ and uniform second moments now imply $\mathsf{F} = A + \widetilde{B} + o_{L^2}(1)$ on compact memory times. Cauchy–Schwarz transfers this replacement to the two-insertion covariance, uniformly in every real external time T, since $X \mapsto e^{iTh} X e^{-iTh}$ is unitary. All four terms $AA, A\widetilde{B}, \widetilde{B}A, \widetilde{B}\widetilde{B}$ remain.

This is a reduction to marked inputs, not yet their spectral solution. The outstanding calculation is the joint law of h, the U_j, their ordered Wick descendants in (32), and the charged defect with its reverse source retained. We need joint-moment formulas at every fixed degree, uniform summation, and control of the zero-frequency spectral mass removed by pinching. Moment convergence alone does not justify passing an exact zero-gap projection through a limit. The present proposition retains the finite-N pinching throughout.

8 Checks, literature, and continuation

An explicit truncation bound. Set $X_S = 2K_*\sqrt{AM}S$ and $C_V = 4\lambda_+\sqrt{2\cdot 10^6}(\sqrt{\lambda_+}+\lambda_+)+2\widehat{L}$. The norm of the jth summand in (40) is at most $C_V e^{2\sqrt{AM}S}(j+1)^2 X_S^j/\sqrt{j!}$. For a cutoff J satisfying

$$r_J = \frac{X_S}{\sqrt{J+2}}\left(\frac{J+3}{J+2}\right)^2 < 1, \tag{49}$$

the tail after J is at most

$$C_V e^{2\sqrt{AM}S}\frac{(J+2)^2 X_S^{J+1}}{\sqrt{(J+1)!}(1-r_J)}. \tag{50}$$

This intentionally loose bound certifies a truncation; it is not an estimate of the unknown joint covariance.

Executed checks. Full-coordinate Hermite tests on $(N,p,m) = (3,1,2),(4,1,2),(4,2,3)$, through degrees $6,5,4$, retain all $9,16,36$ Gaussian coordinates. The 87 identity comparisons include 18 polynomial decompositions and 12 sources; maximum absolute error is $5.68\cdot 10^{-15}$. The nonzero scalar in (45) is explicitly recovered. The rank driver checks 12 full sixth moments at $k=3,4,6$, 38 complete conditional distributions, 350 exact Casimir identities, and 6 independent pure-rank mixtures. Larger-parameter rows give 54 chain and 72 conditional *upper bounds*, not direct deficit moments. All 450 slope-domination checks pass exactly; the largest retained finite relative deviation from the limiting slope is about 0.929. Precision refinement (80/120 digits) changes results by at most $1.95\cdot 10^{-78}$. These checks do not replace the uniform proof. Parameters, dependency versions and hashes are saved.

Existing formulas that can be reused. Effros–Popa [5], Theorems 3.2–3.3, give multiblock q-Wick pairing and product rules. The expected product excludes pairings internal to a block and weights crossings; the product formula also accounts for singletons covered by arcs. This supplies standard combinatorics once the physical rank marks have been identified. It does not identify $\mathfrak{D}$ with a spectral Wick number operator.

Berkooz et al. [6], Sections 4–5 and Appendix C.4, provide uncrossed multiprobe kernels and the crossed four-point R-matrix. Their independent probe weights make source derivatives a possible representation of a marked insertion. Establishing that the derivative equals our common-coupling rank mark is a separate step; no such identification is used in the proposition.

Xu [7], Section 3.4, relates a chord-number generating function to a crossed four-point function. That chord number is distinct from $\mathfrak{D} = p\mathcal{N} - \mathcal{R}$. Direct differentiation of Eq. (101) yields its kth moment as $(-1)^k\partial_\mu^k I|_0$; the extra $1/k!$ printed in Eq. (102) does not follow from that generating-function definition. We use the definition if pursuing this route.

Next calculation. Construct the fixed-degree joint marked contractions first, test their lowest coefficients against the nonzero A_0–B_2 cross measure, and only then sum the controlled series. The same-original-J correlations and precise output-rank marks are the central data. The de Sitter bulk mass/position dictionary and finite-G dynamics remain later tasks. The present result is a microscopic model calculation, with no claim of a complete gravitational dual or of priority over existing formulas. All artifacts remain local.

References

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